

SACHIN GANPULE

US Citizen | (201) 988-5527 | Edison, NJ | [GitHub](#) | [Email](#) | [LinkedIn](#) | [Personal Website](#)

EDUCATION:

University of Michigan, Ann Arbor

Master of Applied Data Science (Concentration in AI)

Expected Graduation: May 2028

Rutgers University, New Brunswick

Bachelor of Science in Electrical & Computer Engineering

May 2025

SKILLS:

Programming: Python (NumPy, Pandas, SciPy, Matplotlib) • SQL • Bash/Zsh • C++ • Java • R

AI/ML & MLOps: PyTorch • HuggingFace • RAG • QLoRA/LoRA • RLHF/DPO • Multi-Agent Systems • OpenAI Agents SDK • LangChain/LangGraph/LangFuse • CrewAI • Docker • Kubernetes/GKE • gVisor • CI/CD • GitHub Actions

Cloud & Dev Tools: AWS (Bedrock, Lambda, SageMaker, EC2, Lambda, S3) • GCP (BigQuery, Pub/Sub) • Git • Claude Code

EXPERIENCE:

AI Intern, Data & Analytics | Volvo Group

Jun 2026 - Present

- Developed a **hybrid RAG pipeline** using **Azure AI Search**, integrating an ingestion layer that processes **multi-format sources**—including portal pages and PDFs—into concurrent Keyword (BM25), Vector, and Document indexes.
- Engineering a time-series forecasting model using **XGBoost** to predict monthly sales target attainment at a **95%** accuracy threshold, delivering data-driven signals to executive leadership for US demand and inventory optimization.
- Improving a CX AI agent by optimizing pipeline latency via a **semantic cache** to avoid recompiling commonly used prompts while integrating **Snowflake Cortex** for advanced data visualization.
- Synthesizing technical tradeoffs and delivering clear recommendations to both engineering and non-technical stakeholders.

AI/Backend Engineering Intern | Hawl Technologies

Jan - May 2026

- Constructed a scalable **pgvector/HNSW** vector search pipeline, replacing TF-IDF and achieving **92% recall@5**.
- Built **Google Drive MCP integration** with OAuth to support content extraction from docs, sheets, slides, PDFs, and text files.
- Improved backend LLM memory infrastructure by developing a batched embedding API (**8x throughput increase**) and migrating rate limiting to Redis, reducing 500-memory backfill time from 8+ minutes to ~1 minute.
- Extended AI Chrome extension to ChatGLM via end-to-end scraper and event handshake for unified conversation capture.
- Standardized development workflows via **Docker** for reproducible testing and **75%** faster engineer onboarding.

Data Analytics Extern | Webacy

Aug - Sep 2024

- Performed EDA, clustering, and risk profiling across 1,000+ smart contracts using **Pandas, SQL, and Matplotlib**. Delivered findings via **Tableau** dashboard and stakeholder presentation.

RESEARCH & PROJECTS:

Latent Delegation in Coding Agent Harnesses

May 2026 - Present

- Spearheading novel research on **latent-space inter-agent delegation** within recursive coding frameworks (RecursiveMAS × LangChain Deep Agents), advancing agent observability to target publication at a **NeurIPS 2026** workshop.
- Designing and executing controlled ablation studies to isolate the efficiency of multi-agent communication, benchmarking critical performance metrics including **success rate, latency, and infrastructure cost**.
- Characterizing the **observability–performance tradeoff**: linear probes at delegation handoffs (file graph, symbols, test deps) plus auditability metrics comparing readable text handoffs to latent bundles.

Autonomous Trading Floor

Mar 2026

- Architected a multi-agent system with **4 autonomous agents, 6 MCP servers, and 44 tools** for live market trades.
- Deployed the system as an enterprise-grade SaaS financial planner using a scalable AWS architecture (**Aurora Serverless, Lambda, SQS**) with **LangFuse** for observability.

GrAldient

Jan - Jun 2025

- Developed an AI-powered grading tool using OCR to automate quiz/test evaluation from PDFs, reducing grading time by **40%**.
- Architected an end-to-end OCR pipeline, transitioning from Tesseract 5 to a fine-tuned **TrOCR** (Vision Transformer) to boost handwriting recognition accuracy from an initial 40% to **97%**. Integrated a **YOLOv8** model for localized text detection and automated preprocessing on **AWS Lambda** using Bash scripts.
- Presented GrAldient at the 2025 Rutgers ECE Capstone Expo; ranked in the **top 15 teams** out of the entire senior class.

CERTIFICATIONS:

[Google Cloud Engineer AI Toolkit](#): May 2026 | [AWS Cloud Practitioner](#): Sep 2025 | [Snowflake GenAI Bootcamp](#): June 2025